[TITLE:]

[SUBTITLE]

by

[Your Full Name]

A Thesis Presented to the
FACULTY OF THE USC DORNSIFE COLLEGE OF LETTERS, ARTS AND SCIENCES
UNIVERSITY OF SOUTHERN CALIFORNIA
In Partial Fulfillment of the
Requirements for the Degree
MASTER OF SCIENCE
(GEOGRAPHIC INFORMATION SCIENCE AND TECHNOLOGY)

[December or May or August] [Year]

Copyright [Year] [Your Full Name]

To your parents, partner, or others you choose

Acknowledgements

I am grateful to my mentor, Professor Smith, for the direction I needed and my other faculty who gave me assistance when I needed it. I would like to thank my employer, X Company, who allowed me to complete some of this work using my office computer. I am grateful for the data provided to me by Y Organization and the assistance in coding the analysis given my friend, Angela Alonzo.

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Abbreviations

GIS	Geographic information system
GIST	Geographic information science and technology
NHC	National Hurricane Center
SGMA	Sustainable Groundwater Management Act
SSI	Spatial Sciences Institute

Abstract

Input your abstract text here. An abstract is a brief summary of your thesis. Do not copy and paste whole sentences from the body of your thesis. Write new text for the Abstract. Only define abbreviations in the Abstract if you refer to the abbreviations in your Abstract. Make sure to end your Abstract with a brief description of your results.

Abstracts are generally only one paragraph, but if you do have a second paragraph in your Abstract, it should be indented whereas the first paragraph is not indented.

Chapter 1 Styles, Headings, and Citations

Welcome to the body of the Spatial Sciences Institute's (SSI) Geographic Information Science and Technology (GIST) Thesis Template. We have stored a number of styles in this document, so please begin writing your thesis in this very document. Click to "Save As", and create a new document. Hang onto this original document, however, as it has helpful tips. Note that this paragraph is not indented. The first paragraph of each chapter, like the Abstract, is flush with the margin. Use the Normal No Indent style.

This is the second paragraph. It is in the Normal style. The bulk of your writing will be in the Normal style. This is not the default Normal that comes with Microsoft Word. We have changed the font and spacing so, again, please use this document rather than a fresh Word document to begin your writing.

All chapters should have some introductory language before you get to the sub-headings. This language should introduce the chapter, including the main sections within that chapter. The introductory text of your first chapter is especially important. You need to clearly state the scope and goal of your thesis project by the time you finish this introductory text.

1.1 First-Level Sub-Heading (Heading 2 Style)

Each chapter should be organized into at least a few sections. Each section begins with a heading in the Heading 2 style. In addition to the proper formatting, the Heading 2 style will provide consecutive numbering in a multi-level list, which includes chapter numbers. You will want to use the Heading 2 style, rather than just mimic the formatting, because Microsoft Word will populate the Table of Contents for your document based upon the text that you have designated with the heading styles and their placement in the multi-level list.

1.1.1 Second-Level Sub-Heading (Heading 3 Style)

Just as each chapter should begin with introductory text, each section and subsection need introductory text before adding any subsections. Thus, Section 1.1 had a bit of text before we reached Section 1.1.1. This text should tell the reader what the content of the section is and what subsections are to come.

1.1.1.1 Third-level sub-heading (Heading 4 style)

Note that the text of this heading was not capitalized. Capitalize the text for first- and second-level sub-headings, but not for third-level sub-heading. You may not require a third-level sub-heading, but if you do, Heading 4 style is here for you.

1.2 Chapters and Headings

The SSI GIST thesis will generally contain five chapters: Introduction, Related Literature, Methods, Results, and Conclusions. Theses that describe a geographic information system (GIS) project developed for online use – a Web GIS project – will sometimes break this up into six, replacing the one Methods chapter, with two chapters that divvy up the topics of user requirements, software selection, data, database creation, and software development, as befits each project.

No section should have only one of the same level of sub-heading. Thus, if you have a section 1.1, you must have a section 1.2 before you move onto section 2. This document breaks this rule above with sections 1.1.1 and 1.1.1.1, but you surely will not make this mistake.

1.3 Citations

The SSI GIST thesis uses the Chicago/Turabian Style of citation formatting, in the author-date style, as it is described by Turabian (2018). As such, you will cite to your sources

using in-text citations, not footnotes. Full citations for references cited in the text will be provided in the back matter, in a list of references.

Chapter 2 Figures, Tables, Captions, and Cross-References

This chapter discusses the inclusion of figures and tables in your thesis document, including how to insert them and how to reference them in the text. The first section covers figures, while the second discusses tables.

2.1 Figures

Introduce and explain all figures with a clear sentence. Explicitly state in the text what is important about the figure. Only share a figure if you have at least a few sentences to say about it. If you do not have this much to say about a figure, it is not important enough to include. Place each figure directly after the paragraph in which you reference it. Do not break up the paragraph. The following paragraph and figure are a clear example.

Figure 1 represents the flood extent analysis for National Hurricane Center (NHC) advisory 15. As shown, observed flooding is predicted well with the two models. A few areas exist where observed flooding was not predicted, but they are isolated west and north of Lake Pontchartrain, along with small pockets towards the coast. Two main areas of difference between the SLOSH predicted extent and that of ADCIRC: the Mississippi River southwest of Lake Pontchartrain and Bayou Lafourche.

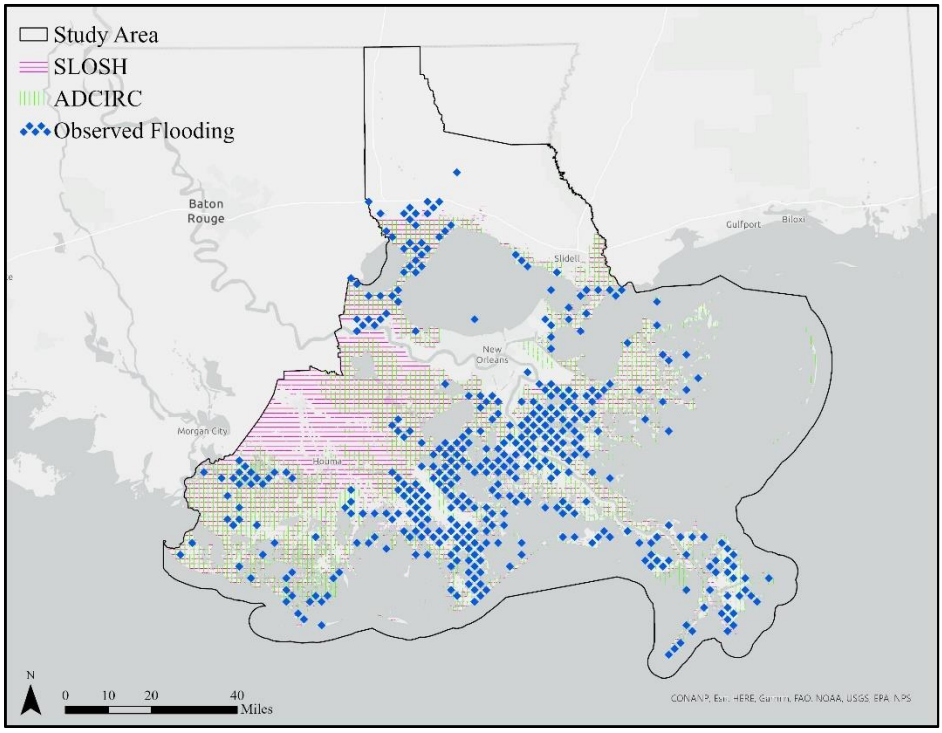

Figure 1. SLOSH and ADCIRC advisory 15 predicted flooding versus observed flooding

You can copy and paste a figure into your document, or you can insert it via the Insert tab. Once you have the figure in the document, apply the Figure style to it. This will center your figure on the page and give it the correct spacing to fall nicely in line with your text.

Each figure needs a caption. To create a caption, right-click on a figure and select “Insert Caption”. The Caption dialog box opens with the self-populated caption of “Figure [#]”. Add a space after the number, and type in a brief caption in sentence-style capitalization. If your figure is a map, this takes the place of a title within the map itself. Save all other description of the figure for the text itself.

The caption serves an important formatting purpose: It identifies the figure as an ordered item within the document. Once you have created the caption, Microsoft Word allows you to link a portion of text to the figure itself (i.e. a cross-reference). If you later decide to reorder your figures (and at some point, you very likely will), the figure numbers in the text and caption will

automatically update. To refer to a figure in the text, place your cursor where you would like the reference to be, then click “Cross-reference” under the Insert tab. Change the reference type to “Figure” and change Insert reference to “Only label and number”. Click Insert and then Close. The text “Figure [#]” will be inserted into the document. If you click on “Figure 1” on page 4 of this document, you will see it has a dark-grey highlight which indicates it is a cross-reference, not just plain text.

Add a space after the caption, and use the Normal Single style for it.

2.2 Tables

Table 1 is an example table that shows the simple formatting expected for thesis tables. Note that there is no shading in the table, and the font is the same as the text. You may use a smaller font size if necessary, but nine point is the smallest size the Graduate School Thesis Center will accept. The table itself is centered on the page. Unlike figures, the table caption is above the table rather than below. Source information, however, is below. The title uses sentence-style capitalization. Add the title for the table following the same instructions above, but choose Table as the reference type and choose Above the Selected Item for the placement.

Table 1. Eastern Bering Sea bottom trawl survey spatiotemporal data

Attribute	Definition	Unit, Resolution	Data Description
YEAR	Year survey conducted	Year, annual	1982 – 2018 surveys
STATION	Station ID	Nominal ID, unique	349 standard shelf stations; 20 x 20 nm stratified grid
LATITUDE, LONGITUDE	Station location	decimal degrees, $1e^{-05}$ °N, °W	Average geographic coordinate location per station (approximate centroid of grid cell)
BOT_DEPTH	Bottom depth	meters, .1 m	Weighted average depth for area swept
BOT_TEMP	Near bottom temperature	degrees Celsius, .1 ℃	Weighted average temperature at maximum depth of trawl headrope
SURF_TEMP	Surface temperature	degrees Celsius, .1 ℃	Temperature measured at surface
CPUE	Catch number per area swept	number/nm^2	Total snow crab; Immature snow crab; Mature female snow crab

Source: Mills 2021

Two table styles have been created: Table Centered and Table Left Justified, for the table header row and content rows, as needed. Try to keep a table entirely on one page. You can prevent a table from breaking across pages by selecting all the text of a table and clicking the "Keep with next" checkbox in the Paragraph formatting dialog box (accessed under the Format tab or via the Line and Paragraph Spacing icon on the Home ribbon). For long tables that necessarily stretch to two pages, repeat the heading row on subsequent pages.

Chapter 3 Document Organization in Microsoft Word

This brief chapter describes just a few points regarding the features you will use in Microsoft Word to organize the various sections of your document.

3.1 Sections and Page Breaks

If you hover your cursor over Break under the Insert tab, you are presented with options for Sections and Page Breaks. Thus far, we have used the word "section" in its general sense to refer to a portion of text organized under a heading. In Microsoft Word parlance, however, a Section is a formally identified portion of a document with different pagination, layout, or header/footers. This document has only two Microsoft Word sections up to this point: one section for the front matter and a second section for the main text. The second section was required to change the pagination at the main text.

As you write your thesis, add a page break at the end of each chapter to move to a new page. Do not add a new section.

Only add a new section if you need to change the page layout or pagination. Since this document contains the necessary sections for pagination of the front matter and text, you should only need to create a new section if you include a large figure or table that requires a page in landscape layout. An example is shown in Table 2, which describes a selection of the San Joaquin Valley's high- and medium-priority groundwater basins as identified by California's Sustainable Groundwater Management Act (SGMA).

Table 2. SGMA basin prioritization for portions of the San Joaquin Valley

Subbasin Name	Basin Area (Acres)	Basin Area (Sq. Mile)	Hydrologic Region	Wells	Wells/sq. mile	Groundwater % of total supply	Basin Priority
Eastern San Joaquin	764802.7772	1195.004339	San Joaquin River	13144	10.99	0.35	High
Modesto	245252.6544	383.2072726	San Joaquin River	4009	10.46	0.37	High
Turlock	348187.0706	544.042298	San Joaquin River	6606	12.14	0.49	High
Merced	512959.0911	801.4985799	San Joaquin River	5892	7.35	0.66	High
Chowchilla	145574.2985	227.4598414	San Joaquin River	1471	6.46	0.96	High
Madera	347667.3941	543.2303033	San Joaquin River	7059	12.99	0.98	High
Delta-Mendota	764964.8592	1195.257593	San Joaquin River	4041	3.38	0.53	High
Kings	981324.8193	1533.32003	Tulare Lake	26684	17.4	0.84	High
Westside	621823.1784	971.5987163	Tulare Lake	1234	1.27	0.82	High
Pleasant Valley	48195.56034	75.30556305	Tulare Lake	87	1.15	0.94	Medium
Kaweah	441003.9175	689.0686211	Tulare Lake	7385	10.71	0.90	High
Tulare Lake	535869.0664	837.2954164	Tulare Lake	3871	4.62	0.50	High
Tule	477646.4035	746.3225055	Tulare Lake	3360	4.5	0.90	High
Kern County	1782320.811	2784.876267	Tulare Lake	6101	2.19	0.79	High

Source: Heap 2022

The page in landscape layout is bracketed by sections of the type “New Section (next page)” They are continuous sections in that the footer formatting and pagination is continuous with the prior sections. This is important as the USC Graduate School requires page numbers in the same location on all pages, be they portrait or landscape layout.

3.2 Front and Back Matter

This section describes the contents and creation of the thesis front and back matter.

3.2.1 Front Matter Contents

The USC Graduate School’s (2022) Thesis and Dissertation Guidelines require the front matter of your thesis to contain a Title Page, Table of Contents, and Abstract. The Dedication and Acknowledgements are optional but are generally included in GIST theses. If your thesis contains any tables or figures, the List of Tables and List of Figures, respectively, are mandatory under the guidelines; if your thesis does not contain any tables or figures (a rare occurrence in a GIST thesis), do not include a List of Tables or List of Figures in your front matter. Include a List of Abbreviations if your document contains more than a handful of abbreviations (quite common in a GIST thesis).

3.2.2 Creating the Front Matter

Create the Dedication, Acknowledgements, Abbreviations, and Abstract by typing the relevant text into the sample pages above, following the placement of the text on the page. We have created the necessary titles for you in this document. If you accidentally delete one of these headings, recreate it using the Heading 1 style and deleting the “Chapter [#]” text.

This template also contains the pages and titles (in Heading 1 style) for the Table of Contents and Lists of Tables and Figures. We have inserted the contents for those three into this

document using Microsoft Word's built-in functionality. This function is found by clicking the Insert tab and then Index and Tables. You can repopulate the Table of Contents to match your headings and repopulate your Lists of Tables and Figures to match your captions.

3.2.3 Back Matter

Your thesis must contain a list of references and may contain one or more appendices (USC Graduate School 2022). Sources that you cited to in your text must be identified in full in the references list. Conversely, if you do not cite to a source in your text, it should not be listed in your references list. Appendices are useful for providing extra documentation about your project. These could include, for instance, extra tables or maps of analytical results, a survey completed by research subjects or testers of a mobile mapping application, or instructions shared with volunteer mappers.

The list of references and each appendix must begin with a title in the Heading 1 style ("References"; "Appendix" (if there is only one appendix) or "Appendix [#]" (if there are multiple appendices)). Page numbers in the back matter continue with the same pagination of the main text, and therefore each only requires a page break to be separated, not a new Microsoft Word section.

References in the list of references must **follow the Turabian style with one key exception: Use only initials for first and middle names**. See the sample list of references below for examples.

References

Heap, C. 2022. "That Sinking Feeling: Predicting Land Subsidence in California's San Joaquin Valley with Spatio-Temporal Regression." Master's thesis, University of Southern California.

Li Sarain, A.Y. 2019. "Providing A New Low-Cost Primary Care Facility for Under-Served Communities: A Site Suitability Analysis for Service Planning Area 6 in Los Angeles County, California" Master's thesis, University of Southern California.

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Turabian, K.L. (2018). A Manual for Writers of Research Papers, Theses, and Dissertations: Chicago Style for Students and Researchers. 9th ed. Revised by W.C. Booth, G.G. Colomb, and J.M. Williams. Chicago: University of Chicago Press.

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